Requirements Documentation

Subgroup 3: Devices

Interactive House Project (Device Subgroup)

Revision History

Date	Version	Description	Author
06/Feb/2026	1.0	Initial skeleton and device list.	Ryad Hazin
25/Feb/2026	1.1	Updated to reflect Keyestudio Arduino hardware kit. removed simulation/GUI requirements.	Ryad Hazin
27/Feb/2026	1.2	Added R6 Serial communication requirement.	Patrick Nordahl
15/Feb/2026	1.3	Added “R7: Command Buffering” and “R8: State-Based Execution”	Tony Nguyen, Patrick Nordahl
21/April/2026	1.4	Updated R7/R8 status to Fulfilled; Added R9 (Authorization),	Tony Nguyen
23/April/2026	1.5	R3: Automated Actuator Response (UPDATED)	Mustafa Al-Bayati, Ryad Hazin, Tony Nguyen, Sergej Macut
08/05/2026	1.6	Updated all files to contain table of content in each file, and references and figures for the files that uses them.	Mustafa Al-Bayati, Ryad Hazin,m Tony Nguyen, Sergej Macut

Requirements List

Requirement Name	Priority
R1. Hardware Initialization	Essential
R2. Real-time Environmental Sensing	Essential
R3. Automated Actuator Response	Essential
R4. Local Visual Feedback (LCD)	Essential
R5. Manual Control (Buttons)	Desirable
R6. Gateway1 Communication (Serial output)	Essential
R7: Command Buffering	~~Essential~~ Fulfilled (Integrated with Middleware)
R8: State-Based Execution	~~Essential~~ Fulfilled (Integrated with Middleware)
R9: Authorization Awareness	Essential

Requirements Descriptions

R1: Hardware Initialization

The system must initialize all digital and analog pins connected to the Keyestudio sensors and actuators (LEDs, Fan, Servo, Buzzer) upon power up.

R2: Real-time Environmental Sensing

The Arduino must continuously monitor analog and digital inputs from the hardware kit, specifically the MQ-2 Gas Sensor, PIR Motion Sensor, and Steam Sensor.

R3: Automated Actuator Response

The system shall implement autonomous behavior, where environmental sensor data triggers automatic actuator responses without requiring user interaction.

Examples include:
If gas level exceeds safety threshold → activate buzzer and fan
If rain is detected → close window + door

This ensures that the system can react in real-time to environmental conditions, even without communication with the Gateway or Server.

R4: Local Visual Feedback (LCD)

Current system data (Light ON, Fan ON, Buzzer!) must be displayed on the LCD Display.

R5: Manual Control (Buttons)

The system should allow manual override of physical components (like the yellow LED or Servo controlled door) using the physical Push Button modules included in the kit.

Note: It is acknowledged that these ***traditional physical interfaces are rigid and potentially inaccessible for individuals with limited mobility****, necessitating the inclusion of remote control through the middleman application.*

R6: Gateway Communication (Serial Output)

Arduino shall be able to transmit sensor values and current device state via Serial communication in a structured format (e.g., JSON or key=value pairs), enabling an external gateway (e.g., laptop or Raspberry Pi) to read and interpret the data.

R7: Command Buffering

Arduino firmware shall implement a non-blocking serial read buffer to ensure that if multiple commands are received in rapid succession from the Gateway, they are executed sequentially without crashing the system

Note: Requirement is primarily handled by the Subgroup 1 Gateway, which serializes incoming commands before they reach the Arduino. The local Arduino buffer now serves as a secondary safety net.

R8: State-Based Execution

Arduino firmware shall check the current hardware state before executing a command to prevent redundant operations and ensure "idempotency" (performing the same action multiple times has the same result as performing it once).

Note: Idempotency is now enforced by the Middleware, which tracks the global state of the house before transmitting a change request to the hardware.

R9: Authorization State Awareness

The system must receive and maintain a real-time "Armed/Locked and away" or "Disarmed/Unlocked and home" state from the central database. This state determines if motion detection triggers a standard convenience light or activates the **Intruder Logic** (flashing LEDs, buzzer alarm, and "DIALING 112" LCD message).